



Product Guide



SATELLITE LEARNING LABORATORY



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Satellite Learning Laboratory Product Guide

Introduction

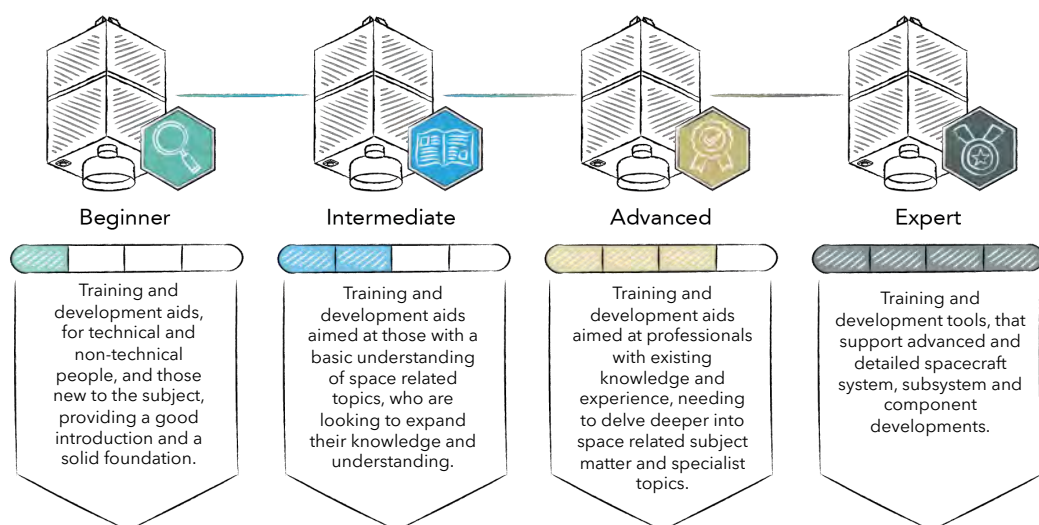
The Satellite Learning Laboratory is a satellite engineering ecosystem that supports workforce development, training and hands-on experimentation, and provides a tool to support spacecraft subsystem, hardware, software and operations procedure development.

The product has been created by spacecraft engineers with years of experience in the design, manufacture, launch and operations of small satellites, and provides an ideal environment for basic, intermediate, advanced and expert users.

The purpose of the Satellite Learning Laboratory is to provide:

1. A tool that can be used to support hands-on satellite related training and workforce development, from beginners, through intermediate, advanced and expert users.
2. A functionally representative model of a satellite that can demonstrate the elements of a satellite system and be used to support payload, and satellite platform subsystem, equipment and software development activities.
3. A tool that provides a bridge between classroom and laboratory use, and the development and operation of satellite flight hardware and software.
4. A tool designed for ease of use, transportation and maintainability.
5. A modular tool that can be developed further to add additional functionality and capabilities or to substitute existing modules and components with alternatives.

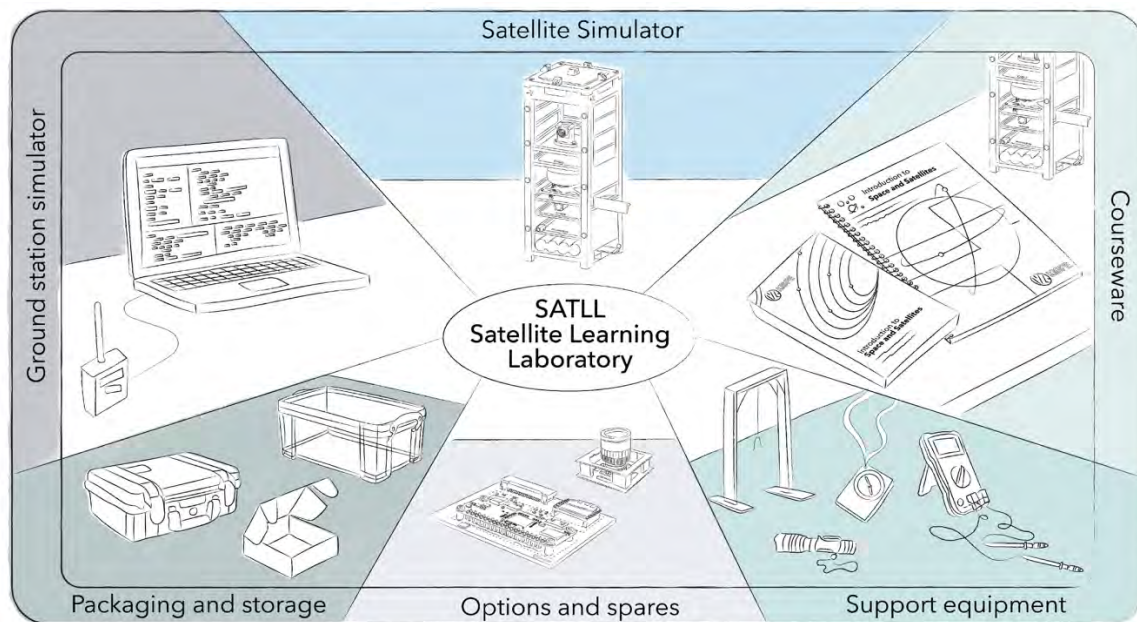
The Satellite Learning Laboratory ecosystem has been developed for use by beginner, intermediate, advanced and expert users, bridging between classroom activities and spacecraft development and operation.



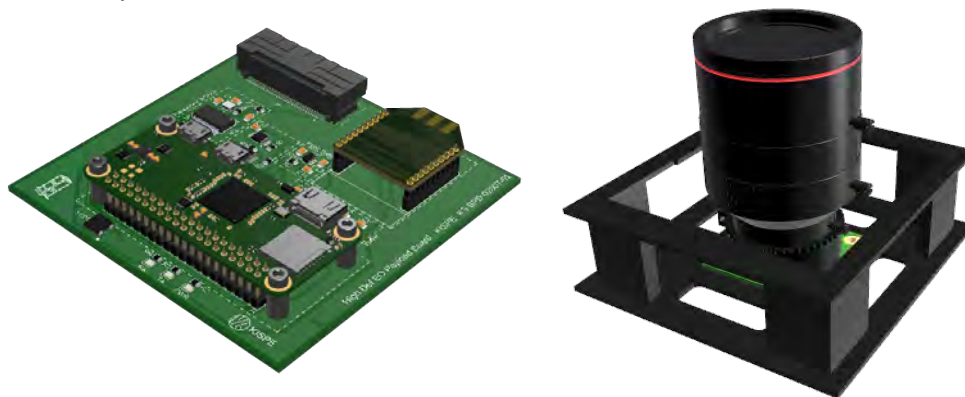
Satellite Learning Laboratory ecosystem

The Satellite Learning Laboratory ecosystem can be viewed as comprising six main elements:

- Satellite Simulator
- Ground Station Simulator
- Courseware
- Support equipment
- Packaging and storage
- Options and spares



The Satellite Learning Laboratory ecosystem has been purposely designed to be modular and expandable, allowing future additions that will expand the capabilities of the system. While KISPE plan to continue adding to the ecosystems, the tool also supports user development, enabling users to tailor or add their own capabilities to address their specific needs.



The image above shows the optional raspberry pi-based imaging payload, and enhancement to the Satellite Learning Laboratory Standard Kit.

Satellite Simulator

The Satellite Learning Laboratory Satellite Simulator, or classroom satellite, provides a functionally representative model of a real satellite.

Electronically, the Satellite Simulator provides representative functions and interfaces, although the implementation and part selection is designed for cost and ground use, and not suitable for use in space.

Mechanically, the Satellite Simulator has been designed for ease of access and use, to facilitate classroom and laboratory activities, rather than survival of the rigors of the launch to orbit.

While functionally representative, you'd never fly the Satellite Learning Laboratory.

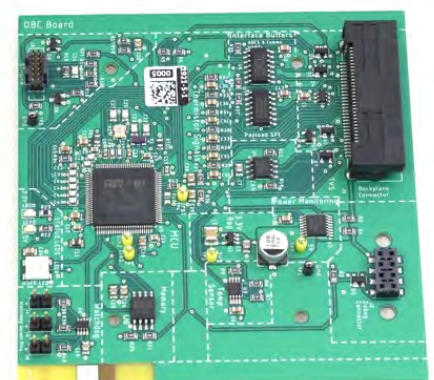
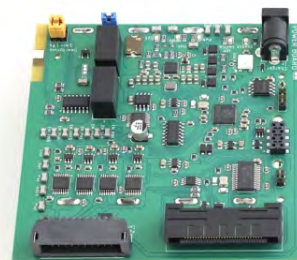
The Satellite Simulator has been designed to be rapidly set up and taken down, maximising the time spent on laboratory activities and experimentation. Subsystems and components can be easily removed and fitted, supporting integration and test-based activities, as well as the replacement or upgrade of modules.



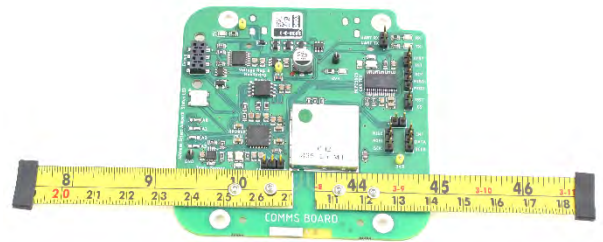
The image above shows the Satellite Simulator fully integrated, and with the subsystems removed.

The key features of the Satellite Simulator provided with the Satellite Learning Laboratory Standard Kit are:

- Launch dispenser (integration and deployment demonstration)
 - The Satellite Learning Laboratory standard kit includes a launch vehicle dispenser, enabling the demonstration of launch integration and deployment.
 - When installed in the dispenser the satellite power is inhibited (spacecraft off).
 - Satellite power comes on after a predefined period after deployment, allowing for sufficient separation distance from the launch vehicle.
- Electrical Power Subsystem (EPS)
 - 7.2v 5000mAh Li-Ion battery
 - Integrated battery charge circuitry
 - Solar array
 - Maximum power point tracking power harvesting
 - Launch vehicle separation switch
 - Remove before flight power inhibit (classroom power switch)
 - Launch deployment power system sequencing
 - Power regulation (5V and 3.3V)
 - Power distribution
 - Power system telemetry (Voltage, Current, Status)
 - Umbilical connection for ground-based power supply during testing.
- On-Board Data Handling Subsystem (OBDH)
 - STM32H573 based computer. The same processor and a subset of the functionality of KISPE's cubesat STM32H573 flight on-board computer.
 - The Satellite Learning Laboratory Standard Kit includes the KISPE STM32H573 Arduino IDE and classroom variant on-board computer software.
 - CAN (Controller Area Network) telemetry and telecommand bus.
 - Distributed telemetry, voltage, current, temperature and other status points.

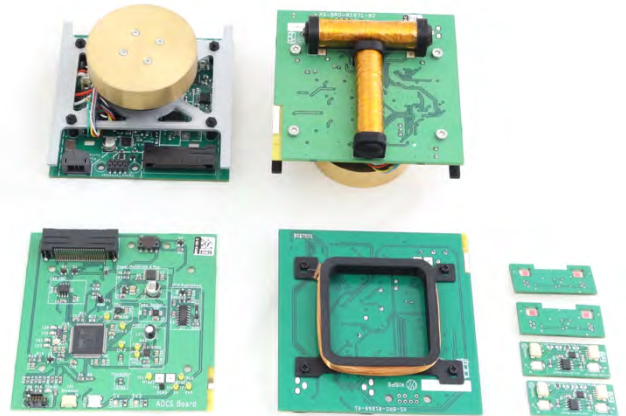


- Communications Subsystem (COMMS)
 - 434MHz (Asia), 868 MHz (Europe) or 915 MHz (US) Ultra High Frequency (UHF) communications half duplex transceiver.
 - UHF band antenna
 - RP2040 based CAN node for communication module telemetry and telecommand.
 - Pre-allocated unit operating frequency, paired with corresponding ground station simulator.
 - Operating frequency and power level modifiable by telecommand.
 - 200kbps default data rate.



- Attitude Determination and Control Subsystem (ADCS)

- STM32 based ADCS computer.
- MEMS Gyroscope
- MEMS accelerometer (not typically used in a functional satellite, but useful for demonstrating spacecraft axis.)
- Deployable magnetometer
- Sun Sensors
- 3 axis Magnetorquer
- Brushless DC motor-based reaction wheel
- Control modes
 - Detumble
 - Sun tracking
 - Manual
 - Idle

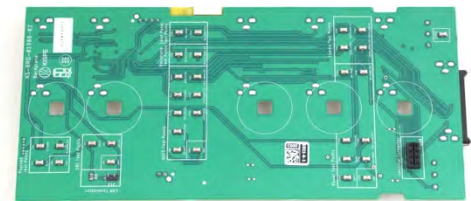
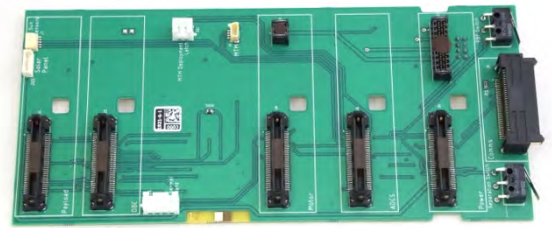


- Thermal subsystem (THERM)
 - Subsystem temperature sensors
 - Thermal experiment panel
 - Heat pipe experiment
 - Thermal surface experiment
 - Thermal control experiment



- Backplane harness

- Integrated backplane providing the harness connections between the various subsystems within the Satellite Simulator.
- Test points
- Umbilical connector for external connection to power and the CAN communications bus.



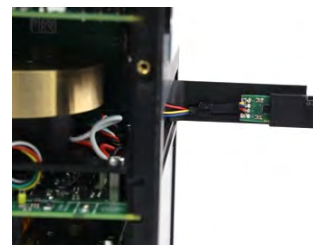
- Structure

- Approximately 100 x 100 x 265mm structure (excluding mounting points).
- Structure designed for ease of classroom experimentation.
- 3D printed structure.
- Acrylic subsystem covers.
- Subsystem alignment rails and removal buttons.
- Removable avionics bay front cover
- Removable payload bay front and back covers.
- Removable payload bay top cover.



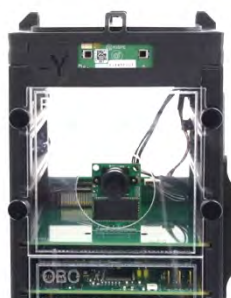
- Mechanisms

- Deployable magnetorquer
- Deployable “Tape Measure” UHF antenna



- Payload

- Modular payload bay (80mm x 80mm x 80mm)
- CAN (controller area network) platform interface
- 5v switched power supply
- Standard payload - The Satellite Learning Laboratory standard kit includes the OV2620 2MP ArduCam imaging payload



Ground Station Simulator

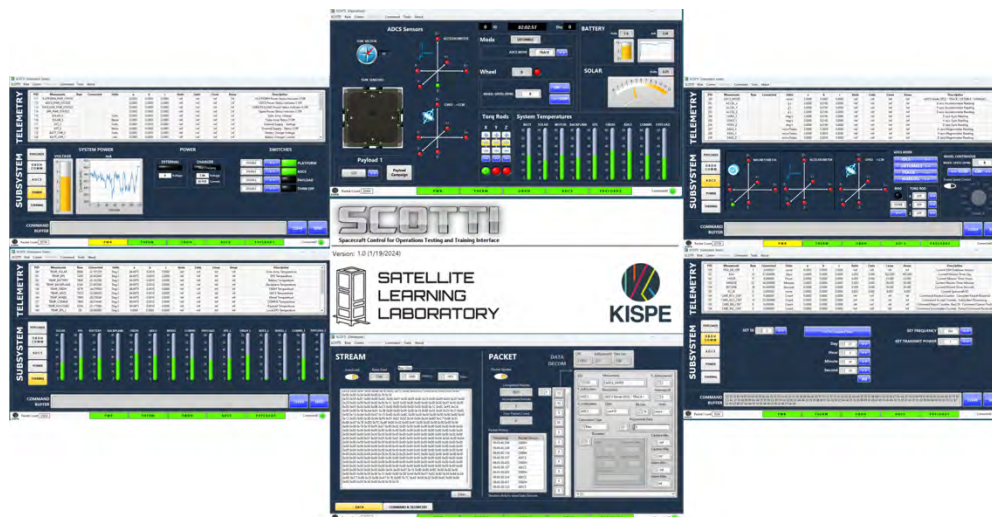
The Satellite Learning Laboratory includes a Ground Station Simulator for communications with the Satellite Simulator and to provide satellite operator or user interface.

The Ground Station Simulator comprises of a user provided computer on which the satellite operations software is installed, and a ground station radio which enables two-way communications between the computer and the Satellite Simulator.



The mission operations software provided in the Satellite Learning Laboratory Standard Kit is a LabVIEW based application, SCOTTI (Spacecraft Control for Operations Testing and Training Interface). SCOTTI was developed initially to support another product, EssentialSAT, used for several years to support KISPE and TSTI's training and workforce development activities. SCOTTI forms the baseline Ground Station Simulator software, ideally suited for beginner and intermediate users.

The software has a range of different displays to be used for different areas of development and learning. Such as simple telemetry interfaces, operational screens, to actual data streams and packet data for more advanced learners.



The Satellite Learning Laboratory Standard Kit is supplied without a computer, with the Ground Station Simulator software supplied on a USB stick.

Courseware

The Satellite Learning Laboratory Standard Kit is supplied with a copy of the “Introduction to Space and Satellites” workbook, intended as an introduction to how satellites work, and the role of the various subsystems and components.

The workbook was developed initially as part of a one-day course for a non-technical audience and those new to the subject, providing a basic understanding of our uses and applications of space, space science, orbits, the purpose of satellites and how they work. The way the experiments are organised, the workbook also provides a good mechanism to familiarise users with the Satellite Learning Laboratory.



The Satellite Learning Laboratory Satellite Simulator enables demonstration and experimentation with all the main satellite subsystems:

- Imaging Payload
- Electrical Power Subsystem (EPS)
- On-Board Data Handling subsystem (OBDH)
- Communications Subsystem (Comms)
- Attitude Determination and Control Subsystem (ADCS)
- Deployment mechanisms
- Thermal
- Structure (it should be noted that the structure is not representative of a flight spacecraft)
- Launch integration and deployment

The Satellite Simulator also enables the demonstration and experimentation in various phases of the satellite concept of operations, from launch operations, through commissioning and then into routine operations.

The Satellite Learning Laboratory has a launch dispenser that inhibits the satellites power system once integrated, a common operating process for cubesat and small satellite missions. There are also several remove before flight items that are removed once integrated into the dispenser, including a remove before flight power inhibitor that can be used as a power switch to turn on and off the unit during classroom activities.

The Satellite Learning Laboratory has a sequenced power system. As the satellite is deployed from the launch dispenser a timer is activated. Once the timer period has been reached, the core satellite subsystems power up, placing the satellite in what would be its safe mode. The timer inhibit of the power system is representative of what many satellites do when coming off the launch vehicle, ensuring that they remain off until they are at a distance from the launcher and cannot cause interference.

The power system then enables the power distribution system to turn on power to the attitude control system so that the satellite can be detumbled and stabilised. The power system also controls power to the payload, turning on the payload for operations, and then off when not in use, again another standard concept of operations for small satellites.

The Satellite Learning Laboratory On-Board Data Handling subsystem provides the platform telemetry, and telecommand functionality, receiving command from the ground station for execution, and collecting and relaying subsystems health status information to the ground.

The Attitude Determination and Control system enables experimentation with sensors such as sun sensors, gyroscopic sensors, magnetometers and an accelerometer. While an accelerometer is not normally used within a real satellite, it is useful for demonstrating the coordinate axis of the satellite.

The Satellite Learning Laboratory includes 3 axis magnetorquers that can be used with a compass to demonstrate the generation of a magnetic field. The unit also includes a single axis reaction wheel, enabling demonstration of satellite detumbling and attitude control modes, such as sun tracking.

Packing and Storage

The Satellite Learning Laboratory Standard Kit is supplied in a rugged storage and transportation case, protecting the contents and providing a robust solution for transportation of the Satellite Learning Laboratory to and from its place of use.

The storage and transportation case has been selected for compliance with cabin luggage compartment requirements of a standard aircraft, maximum dimensions 56cm x 45cm x 25cm.



Support equipment

The Satellite Learning Laboratory Standard Kit is supplied with the essentials required for use, recognising that the users may already have a computer on which to run the ground station software, and test equipment to support operations and use. The additional support equipment required may also depend on the specific use case.

Several optional items are available to enhance the capabilities of the standard kit such as provision of a preinstalled laptop, a basic experimentation kit for those that don't already have the items, and other optional items.

The Satellite Learning Laboratory Standard Kit includes the following support equipment:

- External power supply
 - Multi region input: 100-240V 50/60Hz
 - Output: 12v 2.0A DC.
- Umbilical connector
 - External power connection
 - CAN communications bus interface
- USB Stick
 - SCOTTI Ground Station Software



Customer furnished items

To minimise the cost of the Satellite Learning Laboratory Standard Kit, the assumption has been made that the customer will have their own

computer and the equipment they require to support their use of the Satellite Learning Laboratory.

The Ground Station Software supplied with the Satellite Learning Laboratory Standard Kit has been developed for use on a computer running the Microsoft Windows 11 operating system. A computer with the following specification, or better, is required.



- Intel Core Ultra 5 125U (12 MB cache, 12 cores, 14 threads) processor
- Windows 11 Pro operating system
- 8GB memory
- USB3 interfaces

To complete the exercises within the “Satellite Learning Laboratory – Introduction to Space and Satellites Workbook” the following items will be required.



Multimeter and test leads



500,000 lumen LED flashlight
or Halogen lamp



Strong Neodymium Magnet



Compass



Suspension Frame

Products, options and spares

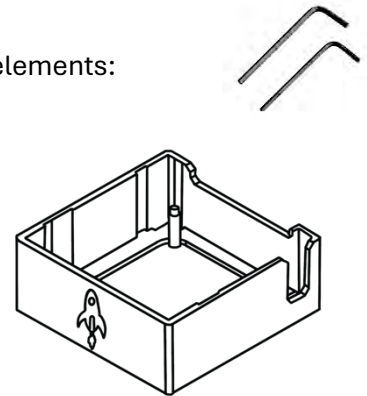
KISPE have developed the Satellite Learning Laboratory to support a wide range of users and use cases, offering a Standard Kit containing the key elements to support the workforce development and training activities, for beginner and intermediate users, and options to expand the capabilities of the baseline Satellite Simulator.

The Satellite Learning Laboratory ecosystem is continuing to be developed with additional capabilities, subsystems, payloads and courseware planned.

Standard Satellite Learning Laboratory Kit

The Satellite Learning Laboratory Standard Kit comprises of the following elements:

- 1x Satellite Learning Laboratory Satellite Simulator
with ArduCam imaging payload.
- 1x Launch Dispenser Simulator.
- 1x Remove before flight power inhibit
- 1x Remove before flight camera cover
- 1x Power adapter
- 1x Ground Station Radio (RF transceiver with USB2 interface)
- 1x USB Stick – SCOTTI ground station software
- 1x Umbilical cable
- 1x Storage and transportation case
- 1x M3 Hexagon Key / Allen Key
- 1x M4 Hexagon Key / Allen Key
- 1x One year warranty
- 1x Quick start guide
- 1x User Guide
- 1x Introduction to space and satellites workbook



The ground station software is for Microsoft Windows 11 operating system.

Option - Satellite Learning Laboratory Suspension Frame

The optional suspension frame can be used to suspend the Satellite Simulator for the Attitude Determination and Control System demonstrations and experimentation, enabling the Satellite Simulator to be able to move around its Z-Axis.

Contents:

- 2x Base struts
- 2x Side struts
- 1x Top strut
- 1x Suspension lines (+spares)
- 1x Satellite Simulator Hanger
- 1x Suspension Frame assembly guide
- 1x Packet of fixing for permanent assembly.



Option - Satellite Learning Laboratory Basic Experimentation Kit

The basic equipment required to support the “Introduction to Space and Satellites workbook” activities.

Contents:

- 1x 500,000 Lumen LED Torch
- 1x Multimeter and test probes
- 1x Magnetic Compass
- 1x Strong Neodymium Magnet
- 1x Storage container



*Make and model of equipment may vary depending on supply

Satellite Learning Laboratory Combo Kit

A kit comprising

- Satellite Learning Laboratory Standard Kit
- Satellite Learning Laboratory – Suspension Frame
- Satellite Learning Laboratory – Basic Experimentation Kit

All packages in a single rugged storage and transportation case.

Option - Satellite Learning Laboratory Laptop with preinstalled ground station software

The Satellite Learning Laboratory Standard Kit includes a USB stick with the windows-based SCOTTI software that can be installed in a user supplied laptop. An option exists for provision of a laptop with the SCOTTI software preinstalled.

Contents:

- 1x Laptop (minimum spec)
 - Intel Core Ultra 5 125U processor
 - 8GB memory
 - USB 2.0 port
 - Windows 11 professional operating system
- 1x Software installation
 - SCOTTI ground station software
 - FTDI device driver
 - Labview drivers



*Make and model of equipment may vary depending on supply

Option – Satellite Learning Laboratory Experiment workbook

An option for provision of ten (10) A4 colour copies of the Introduction to Space and Satellites workbook.

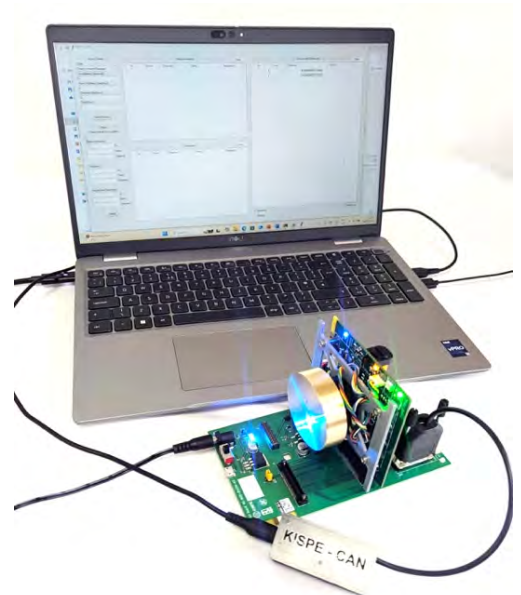
Option – Satellite Learning Laboratory Subsystem Bench Test Module

The subsystem bench test module allows testing of individual subsystems, the Electrical Power System (EPS), On-Board Data Handling System (OBDH), Communications System (COMMS), Wheel Assembly, Attitude Determination and Control System (ADCS) and the Payload.

Power is provided via the power adapter included in the Satellite Learning Laboratory Standard Kit. Subsystems are controlled via a CAN interface, with a Microsoft Windows based application.

Contents:

- 1x Bench test board
- 1x CAN interface module
- 1x CAN interface software
- 1x User guide



Option - Satellite Learning Laboratory HiRes Imaging Payload

A Raspberry Pi based High Resolution Imaging Payload, expanding the capabilities of the Satellite Learning Laboratory Standard Kit, for intermediate, advanced and expert users.

The payload includes:

- A payload controller, allowing power control and monitoring of individual elements of the payload.
- A Raspberry Pi based payload processor.
- An imager based on the Raspberry Pi High Quality Camera, with a 12.3 megapixel, 4056 x 3040 pixel, Sony OMX477 sensor.
- A 6mm Wide Angle Lens.
- S-band payload downlink
- S-band ground station radio
- Payload data reception software.



Option - Satellite Learning Laboratory RTOS and YAMCS ground station software

Per unit software license for enhancement of the Satellite and Ground Station Simulators to use a Real Time Operating System (RTOS) and flight software for the Satellite Simulator, and the YAMCS mission operations software suit for the Ground Station satellite operations.

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